

LE HAI TRUNG

Mechanical Engineering Undergraduate | Robotics, UAV Autonomy, and Soft Robotics

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Education

VinUniversity, College of Engineering and Computer Science

Sep 2023 – Present

Bachelor of Mechanical Engineering | Hanoi, Vietnam

- Cumulative GPA: 3.46/4.00 (as of February 2026).
- Recipient of a 70% merit-based scholarship with an additional 20% financial support; Dean's List, Spring 2025.

Research Interests

Robotics and mechatronics; soft robotics; UAV autonomy; robot perception, mapping, and navigation; control systems.

Research Experience

Undergraduate Research Assistant – Biorobotics Laboratory, VinUniversity

2023 – Present

Supervisor: Dr. Thai Mai Thanh | Undergraduate Research Opportunities Program (UROP)

Teleoperated Humanoid Robotic Hand Using Soft Filament Sensors and Soft Muscles

First Author and Primary System Developer | Manuscript submitted to Journal of Intelligent & Robotic Systems

- Developed the complete teleoperation system integrating lab-developed soft filament sensors, tendon transmission, and hand-fabricated soft-muscle actuation in a 3D-printed humanoid robotic hand.
- Designed and tested the servo-based control architecture in MATLAB/Simulink and evaluated system response using a high-speed camera.
- Demonstrated responsive hand teleoperation with an end-to-end delay below 200 ms.

Hybrid Soft Muscle–Tendon Glove for Post-Stroke Hand Rehabilitation

Co-First Author, System Designer, and Prototype Developer | Manuscript under review at Sensors and Actuators A

- Designed the overall soft robotic exoskeleton glove and fabricated the soft-muscle actuators for assisted finger motion.
- Developed actuators capable of more than 100% elongation and over 2.5 N output force.
- Fabricated and tested the complete prototype, demonstrating assistance for lifting objects up to 350 g in daily-living-type tasks.

Double-Attack Mechanical Thrombectomy Device

First Author and Primary System Developer | Submitted to ICARCV 2026

- Developed a proof-of-concept thrombectomy device inspired by the milli-spinner mechanism, combining rotational clot interaction with rapid axial reciprocation generated by a soft hydraulic muscle.
- Designed a micro-hydraulic turbine mechanism to drive rotation while the soft actuator produced simultaneous push-pull motion.
- Experimentally demonstrated up to approximately 7% improvement in localized suction performance, with greater improvement at lower rotational speeds.

Industry Experience

UAV Engineering Intern – Phenikaa-X, Drone Team

Jul 2025 – Oct 2025

Hanoi, Vietnam

- Designed, 3D-printed, and flight-tested a battery-holder retrofit for the Holybro X500 UAV, ensuring compatibility with the existing platform.
- Implemented and tested PX4 offboard control for GPS-denied flight using AprilTag-based vision localization; conducted circular and figure-eight velocity-control flights.
- Supported onboard camera deployment and data collection and served as a test pilot during flight experiments.

Selected Engineering Projects

Integrated UAV Platform for Indoor SLAM, 3D Reconstruction, and AprilTag

2024 – Present

Precision Landing

Independent Project | Ongoing

- Developed a dual-2D-LiDAR architecture using one horizontal sensor for indoor SLAM and a vertically mounted sensor for environmental scanning and 3D reconstruction in GPS-denied environments.
- Built a reconstruction pipeline that combines sequential 2D scans with UAV pose estimates to generate a stitched 3D point cloud; validated the approach in PX4 SITL and RViz using lawnmower, spiral, and diagonal survey trajectories.
- Deployed the horizontal-LiDAR SLAM pipeline on a custom 10-inch Holybro S500-based UAV equipped with a Raspberry Pi 5 and RPLIDAR sensors.
- Developed and physically demonstrated an indoor AprilTag precision-landing pipeline using local-frame localization, PX4 offboard control, autonomous approach, touchdown, and disarming without GPS.
- Currently integrating the vertical LiDAR, improving point-cloud quality, and extending the physical platform toward complete 3D reconstruction.

Custom Quadcopter Using Teensy 4.1 and MPU6050

2025 – 2026

Group Leader | Intelligent Physical Systems Course Project

- Led the design and construction of an X-frame quadcopter, covering electronics layout, airframe design, integration, and PID-based flight control.
- Built a multi-axis steel-rod-and-bearing test rig and diagnosed vibration-induced IMU errors in the initial 3D-printed frame.
- Redesigned and validated a carbon-fiber airframe using ANSYS, then fabricated, assembled, tuned, and successfully flew the revised platform.

Automatic Tool-Interchange System for a Humanoid Robotic Arm

2025 – 2026

Group Leader | Robotics Course Project

- Led the design of a rotary tool station inspired by CNC tool changers, enabling a humanoid robotic arm to autonomously select and exchange task-specific tools.
- Developed an electromagnetic coupling interface with integrated power-contact pins and implemented the alignment sequence required for reliable mechanical and electrical engagement.
- Successfully demonstrated autonomous tool exchange on the physical robotic system.

Autonomous Whiteboard Pen Refill Machine

2025 – 2026

Main Planner, System Designer, Electrical Engineer, and Fabrication/Assembly Engineer

- Led project planning and overall system design for an automated machine intended to extend marker life and reduce plastic waste.
- Designed and integrated the electrical and motion-control system using stepper motors, drivers, and controllers.
- Designed the CNC-machined acrylic enclosure and served as one of two primary fabricators and assemblers of the functional prototype.

Hand-Gesture-Controlled Mobile Robot for Environmental Surveying

2024 – 2025

Mechatronics Course Project

- Developed a differential-drive robot controlled by hand gestures captured through an IMU-equipped glove.
- Implemented head-motion control of a servo-driven camera gimbal using IMU-equipped glasses for remote visual inspection.
- Integrated ESP32 controllers, RF22 wireless communication, an onboard LCD, and sensors for temperature, humidity, and hazardous-smoke monitoring.

Additional Projects

Wireless Water-Quality Monitoring and Mapping System (pH, TDS, turbidity, temperature, wireless transmission, and Google Maps API); SCARA Robot Kinematics and 2D MATLAB Trajectory Simulation; Buggy Car Chassis Design and Preliminary ANSYS Structural Analysis; 5-Inch ArduPilot Quadcopter; Mini Hawk VTOL Tilt-Rotor Tricopter.

Technical Skills

Robotics and Autonomy: ROS 2, PX4 Autopilot, ArduPilot, MAVLink, Gazebo, RViz, SLAM, offboard control, AprilTag-based localization.

Programming and Control: MATLAB, Simulink, basic Python, PID control.

Embedded and Sensors: Raspberry Pi, Teensy 4.1, ESP32, IMUs, 2D LiDAR, environmental sensors, camera integration.

Mechanical Design and Fabrication: SolidWorks, ANSYS, 3D printing, CNC-machined acrylic, carbon-fiber fabrication, mechanism design, experimental prototyping.

Honors, Admissions, and Certifications

- Admitted to the Robotics, MSE Program at the University of Pennsylvania for Fall 2026 (will not enroll).
- Certified SOLIDWORKS Associate in Mechanical Design (CSWA), Dassault Systèmes, June 2026.
- VinUniversity Dean's List, Spring 2025; 70% merit-based scholarship with an additional 20% financial support, 2023.
- IELTS Academic: 7.5, August 2022.